Borouge plc

Abu Dhabi Securities Exchange · BOROUGE · polyolefins · reports in US dollars, trades in dirhams

Close 2.40 AED · 7 August 2026 · study dated 9 August 2026 · 30.06bn shares in issue

Read this first

What this is. An independent valuation study of Borouge plc, built from the company's own audited and reviewed financial statements and its own disclosed operating data. It is educational analysis. It is not investment advice, it is not a recommendation, and it carries no rating and no price target.

What it gives you instead. A range of fair values from four independent methods, and a separate, explicitly probabilistic map of where the traded price could sit over the next one and three months. The two are different objects and the study never mixes them: the first asks what the business is worth, the second asks what the share price might do.

The one thing to know before reading a single number. This study publishes TWO answers to its central question, not one, and never averages them. Borouge's own share-price history gives a very low measured sensitivity to its market, and a sensitivity built up from what global chemical companies look like gives a much higher one. The gap is worth 1.84 dirhams a share. Rather than pick one and bury the choice, the study carries both all the way through — every lens, every table, the workbook, and the analyst appendix.

Currency. Borouge reports in US dollars. Its shares trade in dirhams, which have been pegged to the dollar at 3.6725 since November 1997 and have not moved since. The valuation is therefore built in dollars and converted to dirhams at the peg; there is no exchange-rate view inside any number here.

Headline

Borouge is a low-cost polyolefin producer in the middle of the worst polyolefin market in a generation, and it is also, right now, a company whose shipping lane was shut for half a year. Those two facts pull in different directions and the study keeps them separate.

On the four valuation lenses the study runs, the field spans 1.29 to 3.33 dirhams a share, with a median of 1.73 against a close of 2.40. That is not a narrow answer and the study does not pretend otherwise. The width is not noise — it is almost entirely one disagreement, about how risky this share really is, and the study prices that disagreement rather than resolving it.

The operating picture underneath is more settled than the valuation range suggests. Borouge earned an EBITDA margin of 37.1% in 2025 while nine of its eleven listed peers lost money. It does that on advantaged ethane feedstock at Ruwais, at a cost of $256 a tonne of production against $395 in the first half of 2026, when the plant had to buy propylene at market prices because its conversion unit was starved of ethane. That gap — about 54% — is the single clearest measure of what the disruption cost.

Three things this study concludes.

One. The 2026 damage is mechanically identifiable and mechanically reversible. Feedstock at $395 a tonne instead of $256, and freight at $177 a tonne instead of $77, are an idled conversion unit and a re-routed ship. Neither is a permanent reset of the cost base. Whether they reverse in 2026 or drag into 2027 is genuinely unknowable, so the study builds both and publishes both: the answers differ by 0.5 fils a share — under a fifth of one per cent of the price — so the disruption is loud in the news and almost silent in the valuation. A year of impaired shipping sits inside a five-year window in which most of the value lies beyond the forecast, so it moves the answer far less than the headlines suggest.

Two. The expansion next door is not yours. Borouge 4 adds 1.4 million tonnes at the same site and is owned 70% by ADNOC and 30% by OMV. Borouge plc operates it for a fee. Valued as a fee stream, that is worth about $2,970m, or 9.7% of enterprise value — real, but a fraction of what the tonnes would be worth if the company owned them. It is why terminal growth in this study is long-run inflation and nothing more.

Three. Most of the value is in the years nobody can see. Terminal value is 72.7% of enterprise value on the central construction. That is high, it is stated in the summary table and in the bridge rather than buried, and it is the honest consequence of valuing a long-lived asset at a cost of capital of 5.48%. At the higher cost of capital the study also publishes, the terminal share falls to 59.7% — and the value falls with it.

Valuation summary

Four lenses, each run twice — once at the share's own measured risk, once at the risk a global chemical company usually carries. The two columns are the same business valued at two different costs of capital. They are never averaged.

Lens	At sector risk (AED)	At own measured risk (AED)	Terminal value as % of enterprise value	What it rests on
Discounted cash flow — shipping normalises	1.49	3.33	72.7%	Five years of tonnes and prices, then a terminal block
Discounted cash flow — disruption persists	1.50	3.32	72.0%	The same build with the lane impaired into 2027
Book value and sustainable return	1.35	2.84	—	Return on equity of 25.3% against the cost of equity
Normalised earnings power	1.29	3.05	—	Mid-cycle volumes and prices, capitalised
Relative multiples	1.73	1.73	—	Three through-cycle multiples; unaffected by the risk debate
The field	**1.29 low**	**3.33 high**		**Median 1.73 against a close of 2.40**

Table 1 — The summary valuation. The terminal-value share is shown against the two cash-flow lenses because that is where it applies; the other three lenses do not have one. Every figure is reproduced live in the companion workbook.

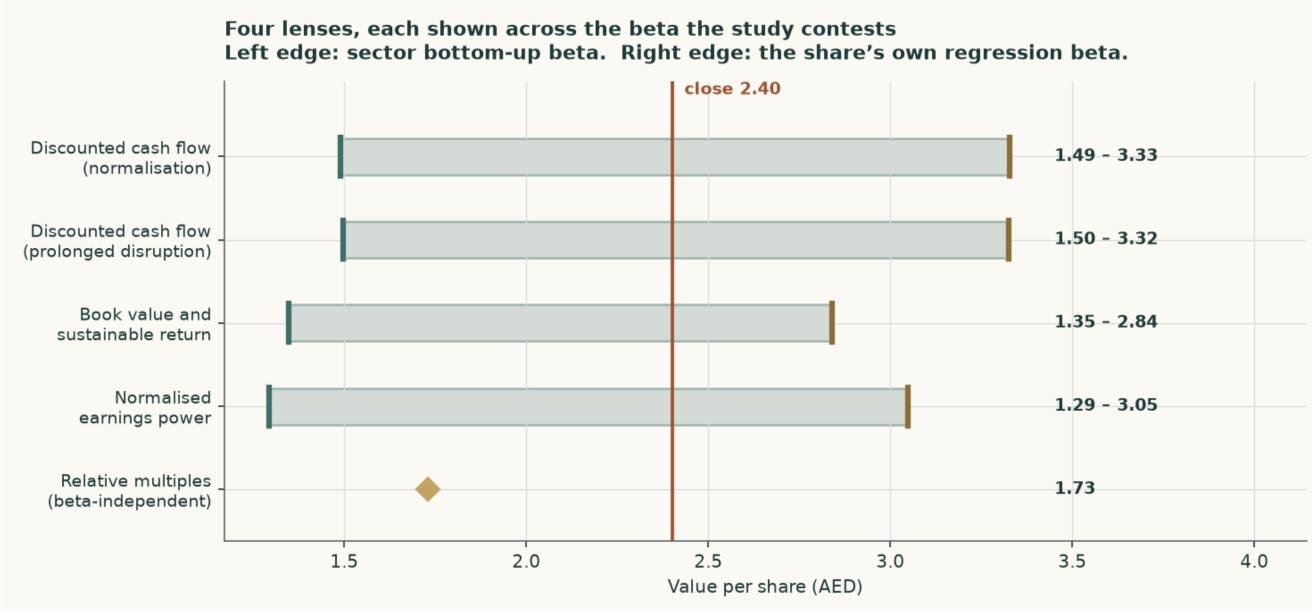

Figure 1 — Each lens is drawn as a bar between the two costs of capital the study publishes. The relative lens is a single point because a through-cycle multiple does not move with the risk debate.

Why the range is this wide, in one paragraph. Take the risk debate out and the study is tight: at the share's own measured risk the four lenses land at 3.33, 2.84, 3.05 and 1.73; at sector risk they land at 1.49, 1.35, 1.29 and 1.73. Within each column the methods agree far better than they disagree. It is the choice BETWEEN columns that is worth almost two dirhams, and no amount of extra modelling settles it — it is a judgement about whether four years of a thinly floated share trading in a narrow band tells you anything about the risk of a business whose earnings track a global commodity.

Company overview

Borouge plc makes two products. It cracks ethane into ethylene and converts it to polyethylene, and it makes polypropylene, at one integrated complex at Ruwais in Abu Dhabi. It sells into more than ninety countries, employs more than 2,900 people, and was formed in 1998 as a partnership between ADNOC and Borealis. It listed on the Abu Dhabi Securities Exchange in June 2022.

Nameplate capacity is 2,750 thousand tonnes a year of polyethylene and 2,230 thousand tonnes of polypropylene. In 2025 the plant produced 5,055 thousand tonnes and sold 5,388 thousand tonnes, the difference being inventory and material sourced from partners. Revenue was $5,848m, EBITDA $2,170m and profit after tax $1,099m.

Two structural facts shape everything that follows, and both were confirmed against the company's own filings rather than inferred.

- **It is now part of a larger group.** Since March 2026 Borouge plc has been majority-held by Borouge Group International AG, formed when ADNOC and OMV combined Borouge and Borealis and acquired Nova Chemicals. Each parent holds 46.94% of that combined entity. Borouge plc remains separately listed and its free float continues to trade. A tender offer converting Borouge plc shares into shares of the combined entity is expected in 2027, subject to market conditions and regulatory approval. No exchange ratio has been published, so no conversion value enters any number in this study.
- **The expansion at its own site belongs to its parents.** Borouge 4, the 1.4 million tonne expansion at the same site, is owned 70% by ADNOC and 30% by OMV. Borouge plc

operates it under an Asset Usage Agreement signed in March 2026 and earns a fee. Its ownership share is zero, and the study carries it at zero. Recontribution of those assets into the parent group is not expected before 2029.

What class of company this is, and therefore how it is valued

The classification matters more than any single assumption, because getting it wrong invalidates the whole study. It was taken from the filings, not from a sector label.

Evidence	What the filings show	What it rules in or out
Revenue mix	Polyethylene $3,436m and polypropylene $2,368m of $5,848m total — 99.3% from selling two polymers	An operating company. No fee income, no rental stream, no lending margin
Balance-sheet shape	Property, plant and equipment $6,082m of total assets $8,489m — 71.6% of the balance sheet is the plant itself	Rules out a holding company and an investment company: there is no portfolio of stakes to value separately
Financial assets	No loan book, no financing receivables, no insurance float. Cash of $427m against borrowings of $2,958m	Rules out a bank and an operating company with a captive lender: there are no legs needing different methods
Segment reporting	One operating segment. Volume and price disclosed by product; cost and profit are not split	A single-lens valuation is correct. A sum-of-the-parts would be inventing divisions the company does not report
The one thing that IS separable	The Borouge 4 operator fee — a contracted stream from assets the company does not own, worth about $2,970m	Valued on its own and added in the bridge, rather than blended into the operating forecast

Table 2 — The classification evidence. Borouge is an operating company with one separable contracted fee stream, so it takes a cash-flow lens built from tonnes and dollars per tonne, with the fee stream valued separately and added in the bridge.

1 Fundamental valuation

1.1 The cash-flow model

The forecast is built from physical units, not from a margin assumption. Volume is nameplate capacity times a utilisation rate the company itself discloses. Price is the published benchmark for each polymer, plus the premium Borouge discloses it earns over that benchmark, times a realisation residual measured against three years of audited revenue. Cost is dollars per tonne, split by what physically drives it.

The realisation residual is the bridge from a published benchmark to the company's own printed top line, and it is measured rather than assumed. Across the three audited years polyethylene revenue per tonne ran 1.0676 times benchmark-plus-premium and polypropylene 1.0321 times. In the first half of 2026 those widened to 1.0947 and 1.0186 on shortage pricing. The forecast carries the audited three-year mean, not the half-year, because the widening is an artefact of a closed shipping lane rather than a durable improvement in what customers will pay.

USD million	2026	2027	2028	2029	2030
Revenue	5,763	5,834	5,632	5,682	5,759
Less: feedstock	1,264	1,332	1,321	1,325	1,332
Less: other production cost	1,554	1,701	1,726	1,741	1,756
Less: selling and distribution	656	504	417	422	432
Less: general and administrative	192	196	200	204	209
Add: other income	27	27	28	29	29

EBITDA	**2,124**	**2,129**	**1,996**	**2,018**	**2,060**
Less: depreciation and amortisation	400	393	389	384	380
EBIT	1,724	1,735	1,608	1,634	1,680
NOPAT (EBIT after tax)	1,228	1,237	1,146	1,165	1,197
Add back: depreciation and amortisation	400	393	389	384	380
Less: capital expenditure	300	320	320	320	320
Less: increase in working capital	11.8	-8.2	-29.3	5.4	8.9
Free cash flow to the firm	**1,317**	**1,318**	**1,244**	**1,223**	**1,248**
Discount factor	0.9480	0.8987	0.8520	0.8077	0.7657
Present value of free cash flow	**1,248**	**1,185**	**1,060**	**988**	**956**

Table 3 — The full free-cash-flow waterfall on the central construction, in the order the reader needs it: EBITDA, depreciation, EBIT, NOPAT, the add-back, capital spend, working capital, free cash flow to the firm, the discount factor and the present value. Every line is a live formula in the companion workbook.

The terminal block	USD million
Final-year NOPAT	1,197
Terminal NOPAT at 2.0% growth	1,221
Reinvestment rate = growth / return on capital of 12%	16.7%
Terminal free cash flow	1,018
Terminal value	29,205
Present value of the terminal value	22,362
Present value of the explicit five years	5,436
Enterprise value of the owned business	27,798
Plus: the Borouge 4 operator fee stream	2,970
Enterprise value	**30,768**
Terminal value as a share of enterprise value	**72.7%**
Less: net debt	(3,275)
Less: lease liabilities	(222.7)
Less: non-controlling interests	(26.8)
Equity value	**27,243**
Value per share (USD)	0.9064
Value per share (AED)	3.33

Table 4 — Enterprise value to equity value, on the central construction. The terminal-value share is shown inside the bridge, where a reader meets it, rather than in a footnote.

1.2 Book value and sustainable return

Borouge earned returns on equity of 21.9%, 27.5%, 26.6% across the three audited years, a mean of 25.3%. Against a cost of equity of 5.74% and long-run growth of 2.0%, that justifies 6.24 times a book value of $0.1240 a share, or 2.84 dirhams. At the higher cost of equity of 9.88% the same return justifies only 2.96 times book, or 1.35 dirhams.

This lens is the most sensitive of the four to the risk debate, and deliberately so: a justified multiple of book is a ratio of two rates, so when the denominator nearly doubles the answer more than halves. It is shown because it is a genuinely independent read on the same business — it uses no forecast at all — not because it is precise.

1.3 Relative multiples

The listed peer set cannot produce an honest earnings multiple today. Of the eleven polyolefin and diversified-chemical peers observed, 9 are loss-making on trailing net income and 2 have no defined enterprise-value-to-EBITDA multiple at all because EBITDA is negative. The median of whatever happens to print is 13.4 times, and it is rejected: a median taken across collapsed denominators measures the depth of the trough, not the multiple the market pays.

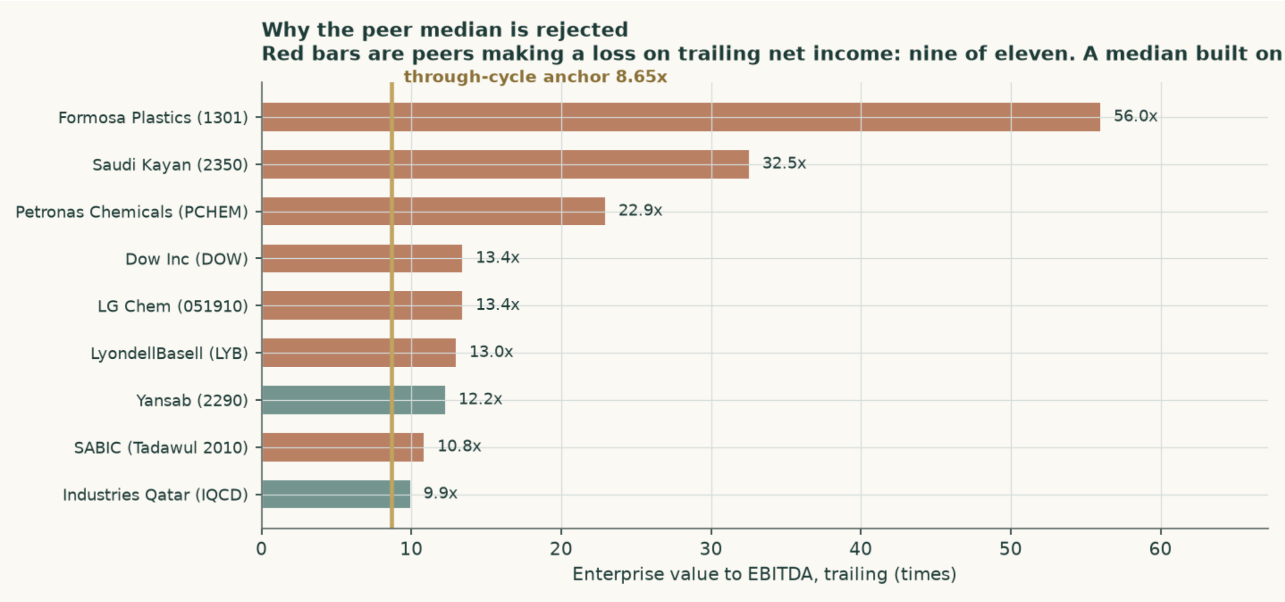

Figure 2 — Nine of eleven listed peers are loss-making. The gold line is the through-cycle multiple the study adopts instead of their median.

Through-cycle anchor	Multiple
LyondellBasell ten-year median EV/EBITDA	7.18x
Industries Qatar current EV/EBITDA	9.90x
Damodaran global Chemical (Diversified) sector EV/EBITDA	8.65x
Median of the three — the multiple adopted	**8.65x**
Applied to mid-cycle EBITDA of $2,042m	$17,671m enterprise value
Less net debt, leases and minorities, per share	**1.73 AED**

Table 5 — The triangulation is shown rather than the conclusion asserted: three named through-cycle anchors, and the median of them. The workbook takes that median in the sheet.

1.4 Normalised earnings power

Mid-cycle is derived from the audited record rather than asserted. Utilisation is the mean of the two most recent audited years — 106.0% for polyethylene and 99.0% for polypropylene — and the benchmark is the mean of the three audited annual averages, $878 and $880 a tonne. Both therefore already contain a turnaround year and a soft-price year, and neither contains the 2026 disruption.

On that basis mid-cycle EBITDA is $2,071m and mid-cycle NOPAT $1,191m. Capitalised at 5.48%, net of the same reinvestment the terminal block charges, that gives 3.05 dirhams a share; at 9.03% it gives 1.29.

1.5 The four lenses in one field

The four lenses across both costs of capital span 1.29 to 3.33 dirhams, median 1.73, against a close of 2.40. Read column by column they agree closely; read across columns they do not. That is the honest shape of the answer and Figure 1 draws it as such.

1.6 The drivers, and what each one is built on

Every forecast driver below is built from something disclosed. Where nothing is disclosed, the study says so rather than inventing a formula.

Driver	How it is built	Source of the build
Volume by product	Nameplate capacity times a disclosed utilisation path, anchored on the rates the plant demonstrated in 2024 and 2025	Management discussion and analysis, by product, every quarter
Realised price	Published benchmark plus the company's own disclosed premium, times a realisation residual measured over three audited years	Management discussion and analysis, reconciled to audited revenue
Feedstock	Split into a contracted ethane leg and a market-priced propylene leg, with the market share of the mix as the driver. Each leg escalates on its OWN price path	Cost per tonne from the audited statements over disclosed tonnes
Other production cost	Fitted across three audited years into a $663m fixed leg and a $201 per tonne variable leg. Only the fixed leg escalates on consumer inflation	Audited statements, three years
Selling and distribution	Dollars per tonne sold. This is the freight route decision, and it is one of the two variables the two constructions differ on	Audited statements and interim statements
Tax	The company's own three-year mean effective rate of 28.7%, not the 9% federal headline — Borouge is taxed under the emirate-level regime	Tax note, three audited years
Working capital	51 days of sales, 62 days of inventory and 92 days of payables, a 20-day cycle. The balance sheet and cash flow are projected from these	Audited statements, three years
Capital expenditure	The current-year guide, then a steady-state maintenance figure set from the company's own three-year outturn. This is the one materially top-down driver in the build, and it is sensitised	Guidance and three years of actual spend
Ethane escalation	Zero real escalation. The pricing formula of the ADNOC feedstock arrangement is not disclosed anywhere, so none is invented	Searched across four annual reports and every filing — not disclosed

Table 6 — The driver table. Margins are an OUTPUT of this build, never an input: the EBITDA margin the forecast produces is whatever tonnes, prices and dollars per tonne make it.

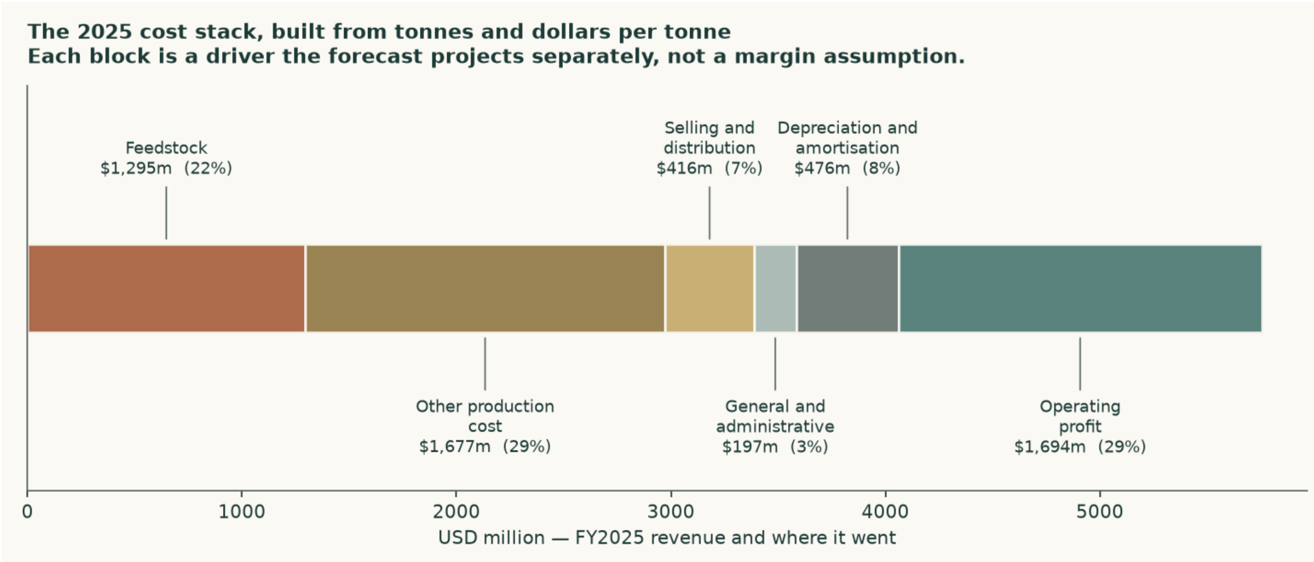

Figure 3 — Where a dollar of 2025 revenue actually went. Each block is a driver the forecast projects separately.

One escalator per cost, and why it matters. The cost stack is not escalated by a single blended inflation index. Contracted ethane escalates on its own terms; purchased propylene escalates on the propylene benchmark path, through the model's own price forecast; only the genuinely domestic fixed leg escalates on UAE consumer inflation of 2.1%. Running one blended index across physically different inputs is the standard way a forecast manufactures a margin trend out of arithmetic — the price side falls on its own commodity path while the cost side rises on a domestic index, and a margin decline appears that nobody chose and no evidence supports.

1.7 The crux

If a reader has time for one number in this study, it is the beta — the measure of how much this share moves with its market, which sets the cost of equity and through it almost the entire valuation range.

	The share's own history	Built up from the sector
Beta	0.271	1.121
How it was measured	216 weekly observations over 5 years against a 17-name UAE market composite	Global chemicals unlevered beta, re-levered at this company's own debt and tax
Statistical quality	R-squared 0.088; 90% interval [0.17, 0.37], spanning 0.72 times the estimate itself	No sampling error of its own, but it is somebody else's companies
Cost of equity	5.74%	9.88%
Cost of capital	5.48%	9.03%
Value per share, shipping normalises	**3.33**	**1.49**

Table 7 — The study's central contested judgement, computed both ways and published side by side. It is never averaged into one number.

The case for the low measured beta is that it is this company's own history, measured properly, and it passes every usability test applied to it. The case against it is that the same tests flag it as weak, and that a lead-lag correction for thin trading moves it DOWN rather than up — so infrequent trading does not explain it. What a 0.27 beta most plausibly measures is a share with a small free float that has traded in a narrow band since listing, not the risk of a business whose earnings track a global commodity benchmark. A reader who finds that argument persuasive should read the sector-beta column; a reader who trusts measured history over analogy should read the other. The study does not choose for them.

1.8 Macro, country and the cost of capital

The cost of capital is built from the ground up, and country risk is charged exactly once.

Component	Value	Where it comes from
US 10-year Treasury yield	4.65%	The dollar risk-free anchor
Less: US default spread	(0.23%)	Normalises the yield to a true risk-free rate
Normalised risk-free rate	**4.42%**	**The rate used**
Dirham cross-check	4.06%	UAE dirham government bond at 4.48% less the UAE default spread of 0.42%
Mature-market equity premium	4.23%	Before country risk
Plus: UAE country risk premium	0.64%	Country risk enters here and nowhere else
Equity risk premium (rating basis)	**4.87%**	**The premium used**
Equity risk premium (spread basis)	4.65%	Published alongside: no sovereign credit-default-swap quote exists for the UAE
Cost of equity — own beta	5.74%	Risk-free plus beta times premium
Cost of equity — sector beta	9.88%	The same build, other beta
Pre-tax cost of debt	5.55%	Marginal and forward-looking, in the currency of the cash flows
Equity weight at market value	85.7%	Market capitalisation $19,643m against net debt $3,275m
Cost of capital — own beta	**5.48%**	
Cost of capital — sector beta	**9.03%**	

Table 8 — The cost of capital, built line by line. The risk-free rate is normalised by the sovereign's own default spread BEFORE a country premium is added, so sovereign risk is charged once rather than twice.

Two constructions of the risk-free rate are shown because they do not agree, and the gap is reported rather than reconciled away. Building it in dollars gives 4.42%; building it from the dirham government bond and subtracting the UAE's own default spread gives 4.06%. Under a hard peg those should coincide; they differ by 0.36%. The dollar construction is used because the company reports, borrows and sells in dollars.

The cost of debt was tested against evidence rather than assumed.

Evidence	Rate	Read
UAE sovereign in dollars	5.07%	The floor — a corporate cannot borrow below its own sovereign in the same currency
Borouge's related-party facilities as priced	5.52%	USD 2.8bn of term facilities plus a drawn revolver from the parent, priced 0.87% over the reference rate
Arm's-length marginal cost adopted	5.55%	The long dollar rate plus a 0.90% margin. The related-party facilities sit 0.03% inside that — close enough to arm's length that no adjustment is warranted, and the test is shown rather than asserted
After tax	3.96%	What enters the cost of capital

Table 9 — The cost-of-debt evidence. Every contested construction in this study is priced rather than argued.

1.9 Sensitivity

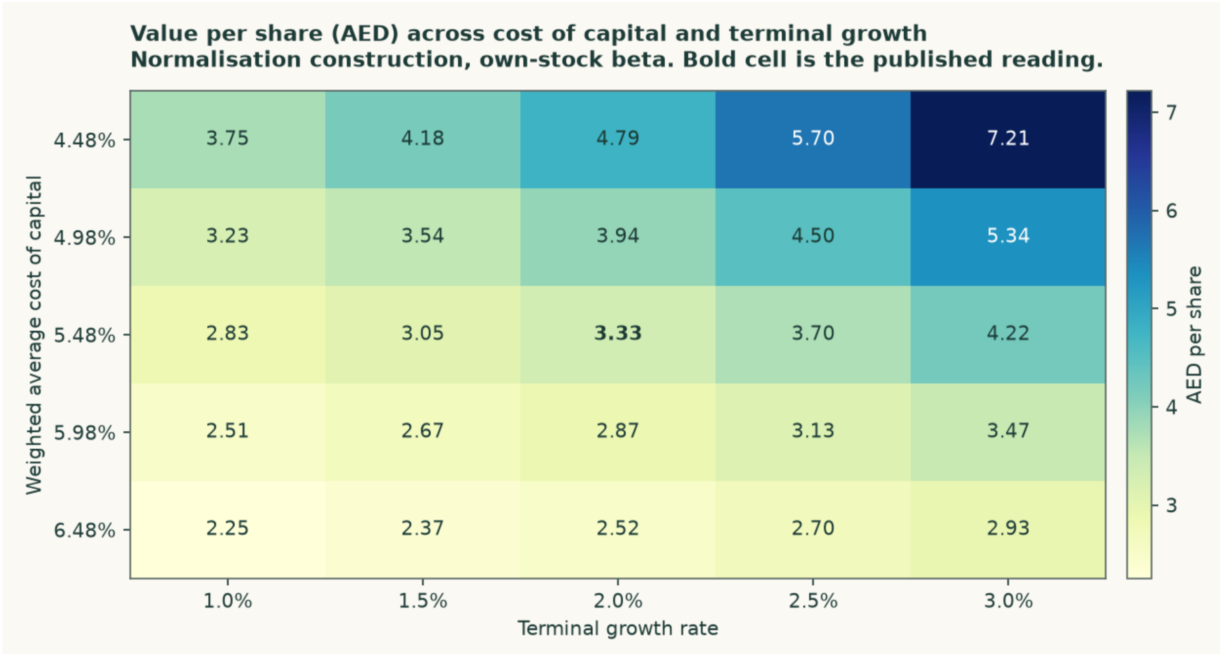

Figure 4 — Value per share across the cost of capital and terminal growth, on the central construction at the share's own measured risk. The bold cell is the published reading.

Two things are worth reading off that grid. The first is how steep it is: half a point on the cost of capital is worth roughly half a dirham, which is why the beta debate dominates this study. The second is subtler — value RISES with terminal growth here, which is the opposite of the usual textbook direction. That is not an error. Growth has to be funded by reinvestment of growth divided by the return on capital, and at a 12% return against a 5.48% cost of capital that reinvestment earns far more than it costs. Had the return sat below the cost of capital the sign would flip.

Sensitivity	Downside	Published	Upside
Realised premium over benchmark, ±$100/tonne	2.16	3.33	4.49
Utilisation, ±10 percentage points	2.69	3.33	3.97
Cost of capital, ±100 basis points	2.52	3.33	4.79

Table 10 — The crux sensitised in observable units: dollars per tonne of premium and percentage points of utilisation, not abstract multiples.

2 Technical and price structure

The share closed at 2.40 dirhams on 7 August 2026, having traded between 2.38 and 2.68 over the past year. It sits 10.4% below that high and 0.8% above that low — which is to say, close to the bottom of its own range.

The price closed 2.40 below a falling 20-day (2.41), a falling 50-day (2.48) and a falling 200-day (2.54). Momentum is soft: RSI(14) is ~39 and the daily ATR near 0.03 (~1.1%) points to an orderly tape. MACD (12·26·9) is below zero but turning up (−0.02 / −0.03 / +0.00). Over the last year it has ranged 2.38–2.68; the last close sits 10% below that high and 1% above that low.

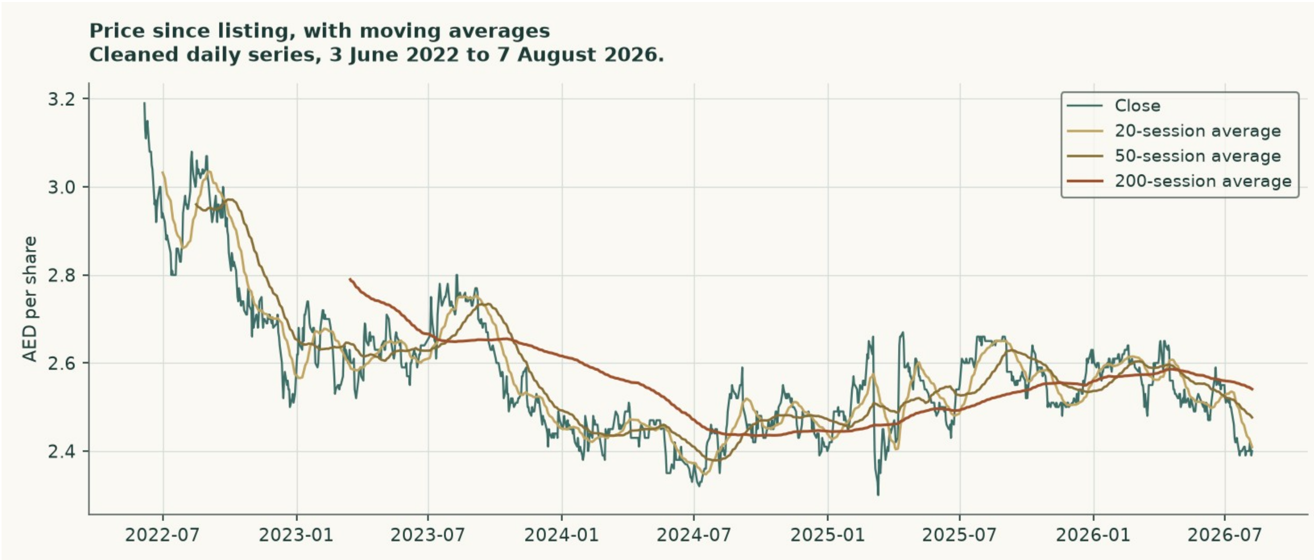

Figure 5 — The price since listing in June 2022, with the twenty-, fifty- and two-hundred-session averages.

Reading	Value	What it says
Close	2.40	The anchor for everything below
20-session average	2.41	Falling — the price is below it
50-session average	2.48	Falling
200-session average	2.54	Falling
Relative strength index (14)	39	Soft, but not at an extreme in either direction
Average true range (14)	0.027	About 1.1% of the price a day — an orderly tape
Moving-average crossover	death cross, 69 sessions ago	Recent enough to still be the governing structure
Nearest resistance	2.43	Then 2.49 and 2.61
Nearest support	2.38	Then 2.29 and 2.20

Table 11 — The price structure. Levels are ordered from nearest to the close outward in both directions.

This section describes the tape and nothing else. It carries no view on what the business is worth — that is what the four lenses above are for — and the two are deliberately not blended.

3 A probabilistic map of the traded price

This is a different question from everything above. The lenses ask what the business is worth. This section asks a narrower and more testable question: given how this share has actually behaved, where could the price sit in one and in three months?

The bands below are simulated from the cleaned daily price history since listing. The central drift is not a forecast of direction — it is the risk-free rate of 3.65% less the dividend yield, which at the company's stated annual intention of 16.2 fils a share on a 2.40 close is 6.75%. Because the dividend exceeds the risk-free rate, the drift is NEGATIVE at -3.10% a year: a share that pays out more than cash earns should, all else equal, drift down by the difference. That is why the probability of finishing above today's close is slightly below half rather than above it.

Horizon	Ends	5th	25th	Median	75th	95th	Above the close
One month	2026-09-07	2.22	2.33	2.39	2.46	2.58	48%

Three months	2026-11-09	2.07	2.26	2.38	2.51	2.74	46%

Table 12 — The percentile map, in dirhams a share.

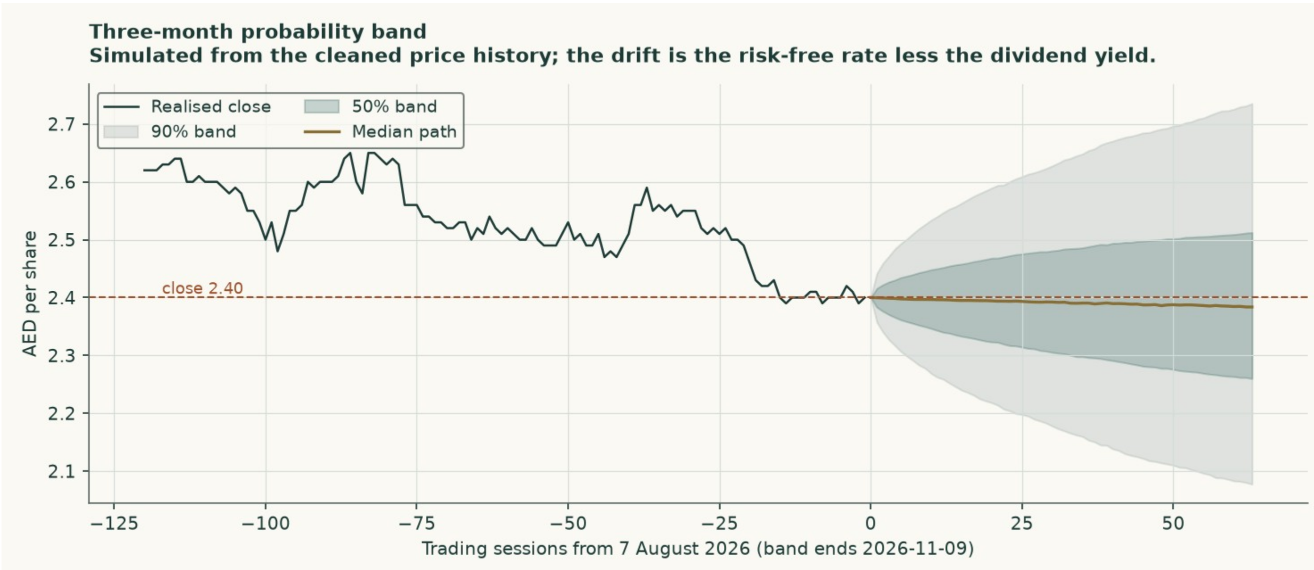

Figure 6 — The three-month band. The shaded areas are the middle 50% and the middle 90% of simulated outcomes.

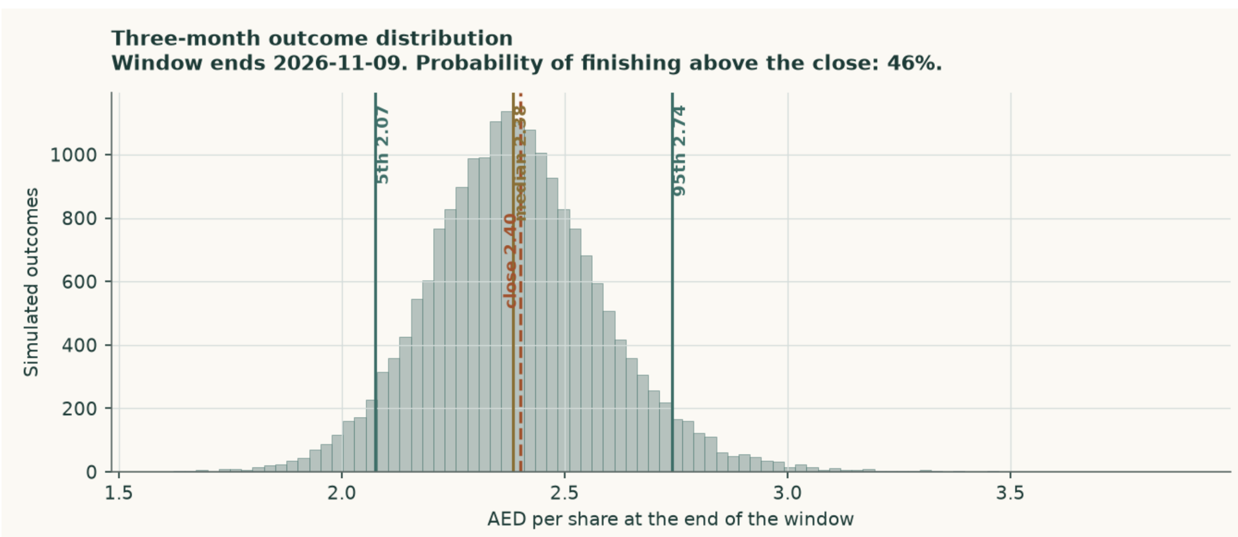

Figure 7 — The full distribution of three-month outcomes.

A level-touch ladder answers a question the percentiles do not: not where the price ends up, but whether it trades through a level at any point along the way.

Move from the close	Level (AED)	Within one month	Within three months
Up 5%	2.52	20%	46%
Up 10%	2.64	3%	20%
Up 15%	2.76	0%	8%
Up 20%	2.88	0%	3%
Down 5%	2.28	21%	50%
Down 10%	2.16	2%	19%
Down 15%	2.04	0%	6%
Down 20%	1.92	0%	2%

Table 13 — The chance the price trades through each level at any point inside the window, not just where it finishes.

How well has this band actually worked on THIS share — and the honest answer. The method was tested by running it backwards over the share's own history, stepping forward three months at a time and scoring each band against what actually happened, against the benchmark of assuming the price simply drifts with carry. On Borouge specifically it did NOT beat that benchmark. Over the 12 independent three-month windows the history allows, it scored -0.090 against it — worse, not better, and worse consistently rather than by luck of the sampling.

What went wrong is specific and it is visible in the numbers. The bands are too WIDE, not mis-centred. 100% of outcomes landed inside the 80% band and 100% inside the 90% band, when 80% and 90% is what those bands are meant to contain. The band came out 1.24 times as wide as the benchmark's. Centring is fine — the average outcome sat at the 0.52 percentile of its own band, close to the midpoint you would want. Borouge is simply a calmer share than the UAE market average the band width is set from, and a band sized for that average is too generous for it.

Two further limits, stated rather than buried. First, Borouge listed on 3 June 2022, so its history is 4.17 years, not five. A five-year test is not available for this share and the five-year window here is identical to its whole history. Second, 12 independent windows is a small sample, and the distribution of outcomes within the bands is not as even as it should be. Read the bands in this section as a description of how this share has behaved, and treat their width as generous rather than tight.

4 Comparison of the lenses

Four methods, deliberately chosen so that they fail in different ways. Where they agree, the agreement means something; where they diverge, the divergence points at a specific assumption rather than at general uncertainty.

Lens	What it is good at	How it fails	Reading (AED)
Discounted cash flow	It is the only lens that uses the actual tonnes, prices and costs, and the only one that can price a fee stream separately	It puts 72.7% of the answer in a terminal block, and it is hostage to the cost of capital	1.49 – 3.33
Book value and sustainable return	Uses no forecast at all — just what the company has actually earned on what it has actually invested	It is a ratio of two rates, so it amplifies the risk debate more than any other lens	1.35 – 2.84
Relative multiples	Completely independent of the risk debate, and anchored on what the market has actually paid through a cycle	The current peer set is broken — nine of eleven are loss-making — so it leans on through-cycle anchors rather than live comparables	1.73
Normalised earnings power	Strips out both the disruption and the trough by construction	Mid-cycle is a judgement, and it inherits the same cost of capital as the cash-flow lens	1.29 – 3.05

Table 14 — What each lens is for, and where each one breaks.

The pattern is clear once the columns are read separately. Inside the low-risk column the four lenses span 1.73 to 3.33; inside the high-risk column they span 1.29 to 1.73. The relative lens sits in the middle of the whole field precisely because it is the one lens the risk debate cannot touch — and at 1.73 against a close of 2.40, it is also the lens that sits furthest below the market.

5 Catalysts

What	When	Why it matters
Third-quarter results	Late October 2026	The first clean read on whether utilisation and freight actually recovered in the second half. This is the single observation that separates the study's two constructions, and it arrives within months rather than years.
Feedstock cost per tonne in the next disclosure	With Q3 and full-year results	It ran $395 in the first half against $256 in 2025. A move back toward the low 300s confirms the conversion unit is running on ethane again; a figure still near $395 says the disruption is structural rather than temporary.
Freight cost per tonne sold	With Q3 and full-year results	$177 against $77 is the re-routing cost. It reverses when the direct route reopens, and not before.
The tender offer	Expected during 2027	Borouge plc shares are expected to convert into shares of the combined group, subject to market conditions and regulatory approval. No exchange ratio has been published. When one is, it becomes the dominant fact about this security and supersedes much of the analysis here.
The dividend	Declared with full-year results	The stated annual intention of 16.2 fils a share is 6.8% on the current price and is explicitly reaffirmed in the half-year release. It is also the reason the probability band in Section 3 drifts down rather than up.
Borouge 4 ramp-up and recontribution	Recontribution not expected before 2029	The operator fee is worth about $2,970m today. If the terms on which those assets eventually move into the group change, so does that figure — in either direction.
Polyolefin benchmark prices	Continuous	Global capacity rises about a fifth by 2034 against demand growing with world output. The industry's own consensus on when operating rates recover has already slipped to 2032. This is the slow variable that sets the ceiling on every lens here.

Table 15 — What to watch, and what each observation would actually settle.

6 Reading the probability zones

The bands in Section 3 are easy to over-read, so this section says plainly what they do and do not mean.

- **A band is a description, not a forecast.** The 90% band for three months runs 2.07 to 2.74. That means roughly nine outcomes in ten fall inside it IF the future resembles this share's own past. It is not a promise, and one outcome in ten falls outside it by design.
- **Finishing and touching are different questions.** The percentile table says where the price might FINISH. The touch ladder says whether it might pass through a level along the way. Those are different probabilities and the touch numbers are always the larger of the two.
- **On this share specifically, the bands are generous.** The evidence in Section 3 says these bands have been too WIDE on this share: 100% of past outcomes landed inside a band meant to hold 90%. A reader should treat the edges as conservative rather than as a tight boundary.
- **The band and the fair value are not the same object.** A fair-value range of 1.29 to 3.33 and a three-month band of 2.07 to 2.74 are answers to different questions. The first is about the business over years; the second is about the share price over weeks. Nothing in this study claims the price will move toward the fair-value range inside the band's window.

7 Caveats, and what would change our mind

The concern	How much it matters	What would settle it

The beta is weak, and it drives everything	It is worth about 1.84 dirhams a share — more than every other uncertainty in this study combined	A longer trading history, a larger free float, or a period in which the share actually moves with its market would resolve it empirically. Until then the study publishes both answers rather than choosing.
Terminal value is 72.7% of enterprise value	High, and structurally so for a long-lived asset at a low cost of capital. At the higher cost of capital it falls to 59.7%	Nothing settles it; it is a property of the method. It is disclosed in the summary table, in the bridge and in the workbook rather than hidden, so a reader can discount it as they see fit.
The ethane pricing formula is not disclosed	Material but bounded. The study carries zero real escalation and shows what a non-zero escalator costs in the sensitivity	Disclosure of the ADNOC supply terms. Until then no forward escalator can honestly be built, and inventing one would be worse than carrying zero and saying so.
Only one operating segment is reported	Volume and price are disclosed by product but cost and profit are not, so the cost allocation between polyethylene and polypropylene rests on physical drivers rather than on disclosure	Segmental profitability disclosure. The gap is flagged rather than papered over.
The tender offer has no published ratio	Potentially decisive, and entirely unquantifiable today. No conversion value enters any number here	Publication of the exchange terms. At that point this study's subject changes and much of it would need re-doing.
Capital expenditure beyond the current year is not guided	The one materially top-down driver in the build. It is set from the company's own three-year outturn and sensitised	A published medium-term capital programme.
The probability bands did not beat their benchmark on this share	Stated in full in Section 3: -0.090 against the benchmark over 12 windows, because the bands are too wide rather than mis-centred	More history. The share has traded for 4.17 years, and the test needs more independent windows than that allows.

Table 16 — What could be wrong, how much it would cost, and what evidence would resolve it.

Appendix A Financial statements

A.1 Income statement — three audited years and five forecast years

USD million	FY2023	FY2024	FY2025	2026	2027	2028	2029	2030
Revenue	5,791	6,026	5,848	5,763	5,834	5,632	5,682	5,759
Gross profit	2,164	2,535	2,282	2,545	2,407	2,197	2,232	2,292
Selling and distribution	399	472	416	656	504	417	422	432
General and administrative	183	193	197	192	196	200	204	209
Operating profit (EBIT)	1,597	1,921	1,694	1,724	1,735	1,608	1,634	1,680
Depreciation and amortisation	572	553	476	400	393	389	384	380
EBITDA	2,169	2,474	2,170	2,124	2,129	1,996	2,018	2,060
Profit before tax	1,401	1,745	1,540	1,542	1,553	1,426	1,453	1,498
Tax	400	506	441	443	446	410	417	431
Profit after tax	1,001	1,239	1,099	1,099	1,107	1,016	1,035	1,068

Table A1 — Historical columns are the audited figures. Forecast columns are the same cells that feed the cash-flow waterfall, linked rather than restated.

A.2 Balance sheet

USD million	FY2023	FY2024	FY2025
Property, plant and equipment	6,677	6,293	6,082
Inventory	645	641	524
Cash and equivalents	354	419	427
Total assets	8,944	8,707	8,489
Borrowings	3,141	2,944	2,958
Equity attributable to owners	4,532	4,463	4,089
Net debt	2,787	2,526	2,531

Table A2 — The audited balance sheet, in USD million, from the filings.

A.3 How the forecast balance sheet and cash flow are built

Neither statement is plugged. The markers below are what the workbook actually does, and a reader can follow each one in the sheet.

Line	How it moves
Property, plant and equipment	Opening balance, plus capital expenditure, less depreciation. Depreciation is the opening balance times a rate set by the company’s own current-year charge.
Trade receivables	Forecast revenue times 51 days over 365 — the collection period the audited statements themselves show.
Inventory	Forecast production cost times 62 days over 365.
Trade payables	Forecast production cost times 92 days over 365.
Net working capital	Receivables plus inventory less payables — a 20-day cash cycle. Its change is the working-capital line in the cash-flow waterfall.
Net debt	Opening balance, less free cash flow after the after-tax finance cost, plus the dividend the company states it intends to keep paying.
Equity	Opening balance, plus profit after tax, less that same dividend.

Cash from operations	EBITDA less the working-capital movement less tax on operating profit. It reconciles to free cash flow to the firm through capital expenditure alone.

Table A3 — The roll-forward markers. Every one of them is a live formula in the companion workbook.

Appendix B Peers, risks and the research record

B.1 The listed peer set

Company	EV/EBITDA	Forward P/E	Price to book	Dividend yield	EBITDA margin	Loss-making
SABIC (Tadawul 2010)	10.8x	24.6x	1.02x	4.3%	13.9%	yes
Saudi Kayan (2350)	32.5x	—	0.99x	0.0%	6.0%	yes
Yansab (2290)	12.2x	33.0x	1.61x	6.7%	20.3%	no
Industries Qatar (IQCD)	9.9x	17.0x	1.86x	6.4%	—	no
LyondellBasell (LYB)	13.0x	7.4x	1.80x	4.6%	7.6%	yes
Dow Inc (DOW)	13.4x	13.7x	1.34x	4.8%	6.6%	yes
Westlake (WLK)	—	20.3x	1.14x	2.7%	-0.7%	yes
Braskem (BAK)	—	—	—	—	-0.1%	yes
Formosa Plastics (1301)	56.0x	14.8x	0.91x	0.9%	1.6%	yes
LG Chem (051910)	13.4x	15.5x	0.43x	0.7%	10.0%	yes
Petronas Chemicals (PCHEM)	22.9x	21.2x	0.97x	1.6%	5.1%	yes
Borouge, on the same basis	**12.0x**	**9.2x**	**—**	**6.8%**	**33.1%**	**no**

Table B1 — The peer set, observed 9 August 2026. It is a cross-check on the valuation, never a source for any Borouge figure. Borouge's EBITDA margin is roughly three times the group's while nine of eleven lose money — the advantaged-feedstock position in one line.

B.2 Risk register

Risk	Where it hits	How the study handles it
Structural oversupply in polyolefins	Benchmark prices in every forecast year	The price path settles BELOW the 2023–24 level rather than returning to it, on the published capacity and operating-rate evidence
Shipping-lane disruption	Utilisation and freight cost	Two full constructions, published side by side rather than averaged
Feedstock availability	Feedstock cost per tonne	The contracted and market-priced legs are modelled separately, with the market share of the mix as an explicit driver
Concentration in one site	Everything	Not diversifiable and not modelled away. One complex, one set of assets, and a single unplanned outage would hit every line
Related-party dependence	Feedstock terms, debt pricing, the fee agreement	The debt is tested against arm's length and sits inside it; the feedstock formula is not disclosed and that gap is flagged
Minority position in a controlled company	Governance, and the terms of any future conversion	Stated as a caveat. No conversion value enters any number
Regulatory pressure on plastics	Long-run demand growth	Carried as a drag on terminal growth rather than as a demand collapse

Table B2 — The risk register.

B.3 The research record

Every historical figure in this study traces to a document Borouge itself published. All 21 of those documents were re-downloaded from the company's own investor-relations library on 9 August 2026 and confirmed byte-for-byte identical to the copies this study was built from. Four complete audited financial years were obtained (2022, 2023, 2024, 2025), of which the three most recent are carried as model columns; both disclosed quarters of the study year were read before the forecast was built, not after. A separate bibliography document lists every source, every input and every judgement.

Layer	What was used	Role
The company’s own filings	Four years of audited statements, two interim periods, five management discussions, four earnings presentations, two earnings releases, four annual reports	The ONLY source of any figure Borouge reports. Historicals, unit build, debt, working capital, tax
Official statistics and central banks	US Treasury yields, the overnight financing rate, the UAE central bank’s base rate and quarterly review, the UAE finance ministry’s bond auction, the IMF and the Energy Information Administration	Macro context and the risk-free rate
Published market data	Sovereign default spreads, equity risk premiums and sector betas by country and industry	The cost-of-capital build
Industry press and research	Capacity outlooks to 2034 and operating-rate consensus	Forecast price direction only — never a Borouge figure
Market data aggregators	Eleven listed peers’ multiples and margins	CROSS-CHECK ONLY, and labelled as such wherever it appears

Table B3 — Where the evidence came from, and what each layer was allowed to do. No aggregator or press report is a source for any figure Borouge itself reports.

Appendix C Three analysts, three methods, one company

Three experienced practitioners were asked to value Borouge independently, each using the method they actually trust. They are labelled Expert 1, 2 and 3. Each shows their workings in full, names a sensitivity with numbers attached, and states in advance the single observation that would prove them wrong.

C.1 Expert 1 — the cash-flow analyst

Worldview. A company is worth the cash it will produce, discounted. Everything else — multiples, book value, sentiment — is a shortcut to that, and shortcuts break exactly when you need them. Build the tonnes, build the dollars per tonne, and let the margin fall out.

When it works: when the operating build is genuinely disclosed, as it is here. When it fails: when most of the answer sits past the forecast, which is precisely the situation — 72.7% of enterprise value is in the terminal block.

Working	Value
Mid-forecast revenue (USD m)	5,632
Mid-forecast EBITDA (USD m)	1,996
Implied EBITDA margin	35.4%
Less depreciation (USD m)	389
EBIT (USD m)	1,608
NOPAT after 28.7% tax (USD m)	1,146
Free cash flow to the firm (USD m)	1,244
Present value of five explicit years (USD m)	5,436
Present value of the terminal block (USD m)	22,362
Operator fee stream (USD m)	2,970
Enterprise value (USD m)	**30,768**
Less net debt, leases and minorities (USD m)	(3,525)
Equity value (USD m)	27,243
Value per share (AED)	**3.33**

Named sensitivity. "Move the cost of capital by fifty basis points and I move by 0.61 dirhams. Move the realised premium by a hundred dollars a tonne and I move by 1.17. Those are the only two that matter to me."

Falsifier, stated in advance. "If full-year 2026 feedstock cost per tonne comes in above $395 — that is, if the first half was not the peak — then my premise that the disruption is temporary is wrong and every number above is too high."

C.2 Expert 2 — the balance-sheet and returns analyst

Worldview. Forecasts are opinions; the balance sheet is a record. What a company has actually earned on what it has actually invested, over a full cycle, tells you more than five years of projected tonnes. Value it off book and off return.

When it works: when a business is capital-intensive with a long record of stable returns, which describes this one. When it fails: when book value has been distorted by write-downs or acquisitions, and when the answer is hostage to the cost of equity — which here it very much is.

Working	Value
Return on equity, 2023	21.9%

Return on equity, 2024	27.5%
Return on equity, 2025	26.6%
Sustainable return on equity (mean)	25.3%
Book value per share (USD)	0.1240
Cost of equity at the share’s own measured risk	5.74%
Long-run growth	2.0%
Justified price to book = (return − growth) / (cost of equity − growth)	6.24x
Value per share (AED)	**2.84**
The same at sector risk: cost of equity	9.88%
The same at sector risk: justified price to book	2.96x
The same at sector risk: value per share (AED)	**1.35**

Named sensitivity. "My lens is a ratio of two rates, so it is the most levered of the three to the risk question — 1.35 against 2.84, a factor of more than two. If the sustainable return falls five points to 20.3%, I lose roughly a fifth of my number."

Falsifier, stated in advance. "If return on equity in 2026 falls below the cost of equity, my whole justification for a premium to book collapses and the right answer is book value itself, not a multiple of it."

C.3 Expert 3 — the cycle and multiple analyst

Worldview. Commodity chemicals is a cycle, and the cycle is the only thing that matters. Do not forecast the trough — normalise through it, and pay a through-cycle multiple on through-cycle earnings.

When it works: at the extremes of a cycle, where forecasts are most wrong and multiples most misleading. When it fails: when the cycle is not a cycle but a structural shift — and with a fifth of global capacity arriving by 2034, that risk is live.

Working	Value
Peers observed	11
Of which loss-making	9
Of which no defined multiple	2
Naive peer median — rejected	13.4x
LyondellBasell ten-year median	7.18x
Industries Qatar, current	9.90x
Global diversified chemicals sector	8.655x
Median of the three — adopted	**8.65x**
Mid-cycle EBITDA (USD m)	2,042
Enterprise value (USD m)	17,671
Equity value (USD m)	14,146
Value per share (AED)	**1.73**
Cross-read: normalised earnings power, own risk (AED)	3.05
Cross-read: normalised earnings power, sector risk (AED)	1.29

Named sensitivity. "One turn on the multiple is worth about 0.25 dirhams a share. That is the whole debate in my lens — and it is why I use three anchors rather than one."

Falsifier, stated in advance. "If global operating rates have not begun to recover by 2029, this is not a cycle and normalising through it is the wrong method. I would switch to a liquidation-and-replacement view of the assets."

C.4 Cross-examination

Challenge	Response
Expert 3 to Expert 1: "Seventy-three per cent of your answer is a terminal block in a business whose industry does not expect to normalise until 2032. You are not valuing cash flows, you are valuing an assumption."	CONCEDED IN PART. The terminal share is high and is disclosed everywhere it appears rather than buried. But the terminal block is built on reinvestment funded at the company's own return on capital, not on a growth assumption pulled from the air — and the sensitivity grid shows what happens across the whole plausible range of both inputs.
Expert 1 to Expert 2: "Your lens has no forecast in it at all, which you call a strength. But you use a cost of equity, and that means you have imported the single most contested input in the study while claiming to avoid judgement."	CONCEDED. The book-value lens is the most levered of the three to the risk question — it moves by more than a factor of two across the two costs of equity. It is presented as an independent read on returns, not as a precise answer, and both readings are published.
Expert 2 to Expert 3: "You reject the peer median because nine of eleven are losing money, then use two of those same companies as through-cycle anchors. You cannot have it both ways."	REJECTED. The rejection is of the CURRENT median, because the denominator has collapsed. The anchors are a ten-year median for one name and a current multiple for a profitable name, plus a sector-wide figure. Those are different objects: one is a trough measurement, the others are through-cycle measurements.
Expert 3 to Expert 2: "A sustainable return on equity of twenty-five per cent in commodity chemicals is not sustainable. It is what advantaged feedstock earns at the top of a cycle, and you have averaged three good years."	CONCEDED IN PART. The three years averaged include 2025, a soft-price year, so the mean is not purely peak. But the point stands that advantaged feedstock is a position, not a law, and the falsifier stated above is precisely the test of it.
Expert 1 to Expert 3: "Your mid-cycle EBITDA is the average of two forecast years from MY model. You are not independent of me — you are a multiple applied to my work."	CONCEDED. The relative lens and the cash-flow lens share a mid-cycle earnings estimate, so they are not fully independent. What the relative lens contributes that the cash-flow lens cannot is a market-tested multiple in place of a discount rate — and that is exactly why it is the one lens the beta debate does not touch.

Table C1 — Each challenge is either conceded or rejected. None is left hanging.

C.5 The three in one room

Put together, the three agree on more than the range suggests. All three accept that the 2026 cost step is mechanically identifiable and probably temporary. All three accept that Borouge 4 belongs to its parents and that terminal growth is therefore inflation and nothing more. All three accept that the peer group is broken as a live comparable set.

Where they part company is the discount rate, and only Expert 3 escapes it. At the share's own measured risk the three land at 3.33, 2.84 and 1.73; at sector risk they land at 1.49, 1.35 and the same 1.73. Expert 3 does not move because a multiple does not contain a discount rate — which is why the study reports the relative lens as the median of the whole field rather than as one reading among four.

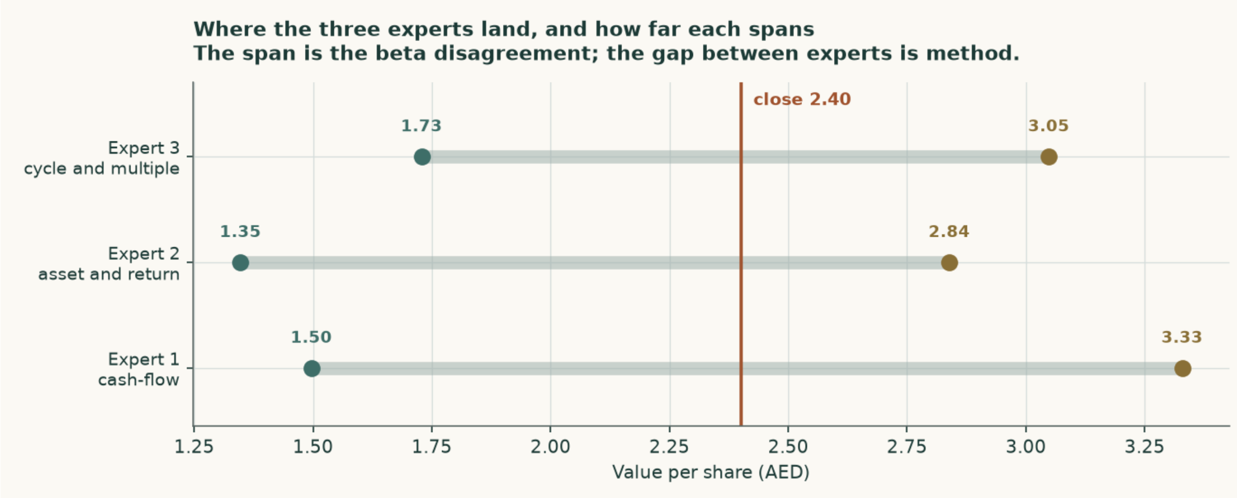

Figure 8 — Where each expert lands. The span within each bar is the risk debate; the distance between bars is method.

C.6 Divergence table — which assumption drives which gap

Gap	Size (AED)	The assumption behind it
Expert 1 at own risk vs Expert 1 at sector risk	1.84	Beta of 0.27 against 1.12 — a cost of capital of 5.48% against 9.03%. This single input is larger than every other disagreement combined
Expert 1 vs Expert 3, at own risk	1.60	A discounted cash flow at a low cost of capital against a through-cycle multiple. Method, not data
Expert 1 vs Expert 2, at own risk	0.49	The forecast adds value above what the current book and return alone justify — the value of the five explicit years
Shipping normalises vs disruption persists	0.5 fils	A year of impaired utilisation and elevated freight, inside a five-year window where most value sits beyond it. Loud in the news, quiet in the arithmetic
Terminal growth 1.0% vs 3.0%	1.38	Return on capital sits above the cost of capital, so growth adds value here rather than destroying it

Table C2 — The divergences, isolated. The beta line is the study's central contested judgement and dwarfs the rest.

About this study

Testahil publishes independent valuation studies and calibrated probability ranges, and keeps a public record of every forecast it makes so that its accuracy can be checked rather than claimed.

Two commitments shape how this document is written. The first is that historical figures come only from the company's own issued statements — never from a data vendor, a broker note or a press summary. Where an official document could not be obtained, the study would say so and stop rather than substitute. In this case every document needed was obtained directly from the company and verified. The second is that a forecast is built from the ground up: product by product, tonnes times price, cost per tonne, with each cost escalating on the thing that actually drives it. Margins are an output of that build, never an input to it.

Where a figure has two legitimate constructions, both are published. This study does that twice — for the cost of capital and for the shipping disruption — and averages neither.

Disclosure

This document is educational analysis and is not investment advice, an offer, or a solicitation. It is not a recommendation to buy, sell or hold any security. It contains no rating and no price target, by design: it publishes ranges and probability distributions instead, because a single number would imply a precision that the evidence does not support.

The analysis rests on information believed to be reliable, drawn from the company's own published filings and from official statistical sources, but its accuracy and completeness are not guaranteed. Valuation involves assumptions about the future that will not be borne out exactly. The probability ranges in Section 3 describe how this share has behaved historically and are explicitly not promises about how it will behave.

All figures are as at 7 August 2026 for market data and the dates stated for financial data. The study was prepared on 9 August 2026. Currency amounts are US dollars for financial statements and dirhams for per-share figures, converted at the pegged rate of 3.6725 dirhams to the dollar.